Table 8: Comparison of STM and regularization strategies on the MBPP Ground Truth data using Llama3 8B Instruct. Hyperparameters are selected based on the performance or a similar L2 norm of ΔW to STM's setup. STM consistently outperforms all regularization methods, suggesting that degradation is driven more by high-PPL training than by overfitting.

Regularization & Hyperparameter	**L2 norm of ΔW**	**BWT (%)**	**TI (%)**
WEIGHT DECAY 0 + DROPOUT 0.05	0.7539	-9.24	-9.82
WEIGHT DECAY 0.2 + DROPOUT 0.3	0.7109	-3.50	-8.03
WEIGHT DECAY 0.5 + DROPOUT 0.3	0.5351	-11.15	2.86
KL $coef$ =1E-5	0	-0.24	2.24
STM	0.5500	**1.90**	**3.12**

models for distillation could be challenging. [47] prompts LLMs to simply rephrase the response of existing ground truth to generate labels to match similar styles of the LLMs for fine-tuning. However, rephrasing the ground truth answer limits the output distribution and results in lower performance in our study. Furthermore, [11] exploits a base LLM as a judge to pick out answerable and unanswerable questions to compose a new training dataset to improve target task performance only on the QA/conversation dataset. Although using Mistral 7B Instruct to generate acceptable responses for answerable questions improves target and non-target tasks, it is rarely discussed that if the proposed method is applicable to different model sizes and series. Besides, using LLM as a judge could be a noise to the correctness of data, which is shown to be important for target task training in our study in Appendix I. Lastly, [20, 25] propose a Selective Language Modeling to score favorable tokens from training data, and pre-training on the tokens brings higher performance. However, the requirements for pre-training a reference model or preparing a high-quality dataset for a scoring model could be exhausting depending on task difficulty. In addition to STM's applicability in Fine-Tuning, STM does not require pre-training for the selection of tokens at all, which brings efficiency and performance improvement at the same time.

8 Conclusion

In this work, we present a comprehensive empirical study of how fine-tuning with LLM-generated data enhances cross-domain generalization in instruction-following models. Our findings reveal that such data significantly reduces degradation on non-target tasks compared to traditional ground truth fine-tuning. This robustness is closely linked to the reduced presence of high perplexity tokens in input sequences, which we identify as a critical factor in preserving general capabilities. Building on this insight, we introduce the Selective Token Masking (STM) method—an efficient and model-agnostic strategy for achieving comparable improvements without relying on external model generations. Our extensive evaluations across diverse models, training strategies, and learning settings underscore the generality and effectiveness of low-perplexity training. These findings offer a loss- and perplexity-based perspective on fine-tuning robustness and point to a practical direction for future work: designing adaptive strategies that prioritize low-perplexity tokens or filter out harmful high-perplexity tokens. Such approaches may generalize across models and tasks, enabling efficient domain adaptation while preserving general capabilities.

9 Limitations

The effectiveness of token-wise masking on training models could relieve catastrophic forgetting and improve target task performance simply. Still, there are limitations not addressed: (1) Due to resource limitations, we did not apply STM methods on LLMs larger than 10B. (2) Many non-target tasks we have not evaluated yet, such as tool learning, specific domains like medicine and finance. (3) Self-output data generation is costly. A trade-off between computing resources and performance can be further studied. (4) We did not consider how perplexity changes when task difficulty scales up. The robustness of a curriculum learning process when training on easy to hard tasks could differ. (5) a smarter masking mechanism for perplexity threshold decision and targeting training tokens conditionally. Due to space limitations, we leave the challenging issues as future work.

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